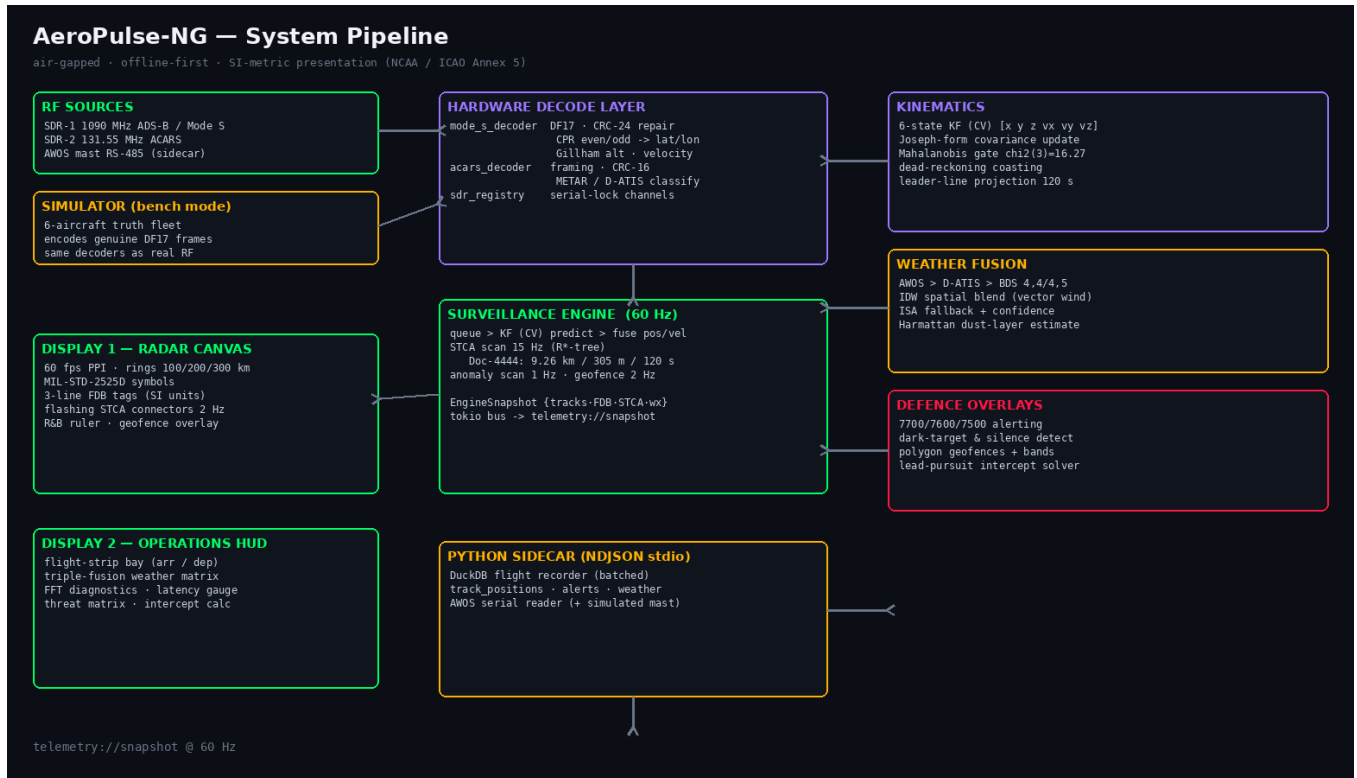


1. System Overview

AeroPulse-NG is an air-gapped surveillance workstation: software-defined radios feed a Rust fusion engine that publishes a 60 Hz tactical snapshot to two operator displays, while a Python sidecar persists flight history and reads the AWOS weather mast. There is no internet dependency anywhere in the pipeline.



1.1 Guiding constraints

- Air-gapped by contract — no build-time or runtime network access; vector-drawn PPI instead of map tiles.
- One decode path — synthetic traffic is encoded into genuine DF17 frames (CRC parity included) and pushed through the production decoder.
- Deterministic core, async shell — the engine is a pure state machine driven by explicit timestamps; the Tauri runtime owns clocks and I/O.
- Presentation converts once — engine internals stay native (metres, m/s; wire ft/kt per DO-260B); SI formatting happens only at the display boundary.

2. Ingestion & Decoding

Three receiver channels bind to factory serial numbers (SDR-1 ADS-B 1090 MHz, SDR-2 ACARS 131.55 MHz, SDR-3 VHF guard). The Mode S decoder validates CRC-24 parity (remainder equals ICAO address, or zero for even parity), repairs single-bit errors across all 112 positions, and extracts identity, position, velocity and aircraft status. Position resolves through Compact Position Reporting: an even/odd frame pair inside a 10-second epoch yields absolute latitude/longitude via the standard zone arithmetic (dlat $6^\circ / 360/59^\circ$, NL-table guard, longitude index m). Altitude decodes the 12-bit field with the Q-bit arithmetic path or legacy Gillham gray code. ACARS frames parse SOH..ETX structure with CCITT-CRC integrity; weather-bearing labels classify into METAR/SPECI and D-ATIS products.

2.1 The synthetic feed

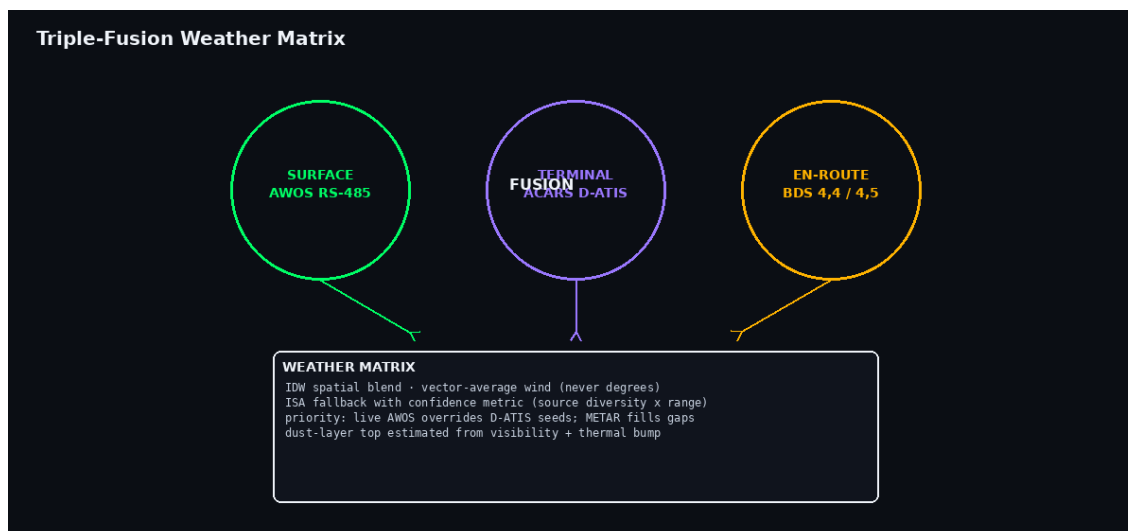
The bench simulator advances a six-aircraft truth fleet and encodes every observation into genuine DF17 frames — CRC address parity included — before handing them to the same decoder. Deterministic seeding (xorshift64*) makes the frame timeline bit-reproducible. Scripted events: emergency squawk at T+45 s, an analytically solved intercept crossing VL604's path at T+70 s with a sub-kilometre predicted miss, scheduled RF dropouts, and a dark target that never broadcasts identity.

3. Kinematics

Each track carries a 6-state Kalman filter (constant-velocity, linear; module ekf.rs historic) [x, y, z, vx, vy, vz] on the local East-North-Up tangent plane. The process model is piecewise-constant white-noise acceleration; covariance updates use the Joseph form for numerical stability; a Mahalanobis gate (chi-squared, 3 dof, 16.27) engages after a 12-fix acquisition warm-up — before the velocity state converges, constant-velocity prediction errors legitimately exceed steady-state bounds. During signal loss the filter simply keeps predicting: dead reckoning falls out of the same equations, and the display flags coasting tracks with dashed halos.

Conflict detection bulk-loads every track's 120-second predicted corridor into an R*-tree (Beckmann split criteria with forced reinsertion), retrieves candidate pairs by rectangle query, then refines with 5-second time-stepping that requires simultaneous infringement of the Doc 4444 minima — 9.26 km horizontal and 305 m vertical in the SI presentation, evaluated internally as the regulatory 5 NM / 1,000 ft constants.

4. Triple-Fusion Weather



Surface truth arrives from the AWOS mast through the Python sidecar; terminal-sector text arrives over VHF as ACARS D-ATIS/METAR; en-route profiles come from Mode S BDS 4,4 (wind) and 4,5 (temperature deviation) register downlinks. Airborne samples accumulate in a bounded deque (15-minute TTL) and blend by inverse-distance weighting with an altitude-mismatch penalty — wind averages as vectors, never by averaging degrees. Pressure blends toward the ICAO standard atmosphere in proportion to altitude mismatch; every query returns a confidence score from source diversity and range. A Harmattan dust-layer estimate rises from visibility degradation plus a convective thermal bump above 28 °C surface temperature.

5. Defence Overlays

- Squawk semantics: 7700 general emergency, 7600 radio failure, 7500 unlawful interference — mapped to alert states with display priority.
- Dark targets: identity-less emitters flag after a 30 s arming window; previously cooperative tracks that go silent flag after 45 s.
- Geofencing: ray-cast polygon inclusion with floor/ceiling bands; entry events only, 30 s hold-down, exit re-arms.
- Intercept solving: damped fixed-point pursuit iteration with forward-simulation verification; minimum feasible speed by bisection.

6. Host Integration

The Tauri runtime owns the clock: a 60 Hz tokio task ticks the simulator, drains queued events through the engine, and emits telemetry://snapshot to both windows. Twelve IPC commands cover snapshot pulls, status, weather sampling, geofence administration, emergency override and intercept computation. The sidecar speaks NDJSON over stdio (init / log_tracks / shutdown; ready / awos / db_ack) and degrades gracefully — losing persistence never takes surveillance down.

7. Key Decisions & Trade-offs

Decision	Rationale	Rejected alternative
Hand-rolled 6x6 linear algebra	Zero transitive native deps; bit-exact reproduci	nalgebra (audit surface)
Bulk R*-tree rebuild per pass	Tens of microseconds at operational scale; simpl	k-d tree / gridbalancing
Joseph-form covariance	Stability across the wide dynamic range of acqui	(I-KH)P form
Queue-then-drain ingestion	Fixes double-prediction: measurements fuse ag	fuse-inside-ingestexactly once
JSON IPC at 60 Hz	Webview simplicity; ~40 KB/frame measured; z	SharedArrayBuffer ring ed
Canvas2D renderer	60 fps sustained at operational track counts; nc	Pixi/WebGL now

8. Roadmap

- RTL-SDR I/Q front-end: libusb transfers, DC/PPM correction, Manchester sync feeding the existing decoder.
- Live FFT spectrum from the RF tap (plot currently synthesised from message-rate telemetry).
- Physical AWOS bring-up and Comm-B interrogation scheduling for live BDS weather.
- Alert/weather persistence writers; zero-copy IPC migration; multi-site handover and RBAC views.